

JOONWON LEE

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SUMMARY

Statistics Ph.D. candidate and quantitative researcher specializing in time-series and spatiotemporal modeling, stochastic processes, scalable likelihood inference, and statistical risk analysis. Develops Python/PyTorch models and simulation-based diagnostics for large, noisy datasets and translates quantitative results into decision-relevant risk insights.

EDUCATION

Ph.D. in Statistics, Rutgers University, Piscataway, NJ Expected Dec 2026 | GPA: 3.8/4.0
M.S. in Statistics, University of Minnesota, Minneapolis, MN Sep 2019 - Sep 2021 | GPA: 3.7/4.0

TECHNICAL SKILLS

Programming and Computing: Python (NumPy, Pandas, SciPy, PyTorch, LightGBM, scikit-learn), SQL/MySQL, R, Linux, Git, AWS EC2, HPC/SLURM

Quantitative Methods: Time-series and spatiotemporal modeling, stochastic processes, MLE/GLS, numerical optimization, Monte Carlo simulation, hypothesis testing, spectral analysis, predictive modeling, model validation, risk analytics

EXPERIENCE

JPMorgan Chase - Summer Quantitative Analytics Associate, Quantitative Finance / Risk Modeling Summer 2026

- Developed and validated a Sequential Probability Ratio Test (SPRT) framework for binary risk indicators, translating a business monitoring problem into statistically controlled decision rules.
- Designed simulation-based evaluation of Type I/II error, detection delay, and early-decision tradeoffs, using Brownian-motion approximations to characterize stopping-time uncertainty.

Travelers - Data Science Leadership Program, Business Insurance Pricing Team Summer 2025

- Developed an end-to-end LightGBM pricing pipeline for mid-market property risk, including data preparation, GLM benchmarking, model validation, and performance diagnostics across 2.46M+ records on AWS EC2.
- Presented model diagnostics and feature-attribution findings as pricing and risk-segmentation recommendations for business stakeholders.

QUANTITATIVE RESEARCH AND TRADING PROJECTS

IMC Prosperity Algorithmic Trading Competition - Top 2.39% of Participants Apr 2025

- Built and evaluated a market-making strategy with inventory-aware quoting, dynamic liquidation rules, and benchmark-driven performance analysis.

Ph.D. Dissertation Research - Scalable Time-Series and Spatiotemporal Modeling Sep 2024 - Present

- Scalable Statistical Modeling:** Developed an advection-aware Vecchia likelihood approximation for nonseparable Gaussian processes, replacing dense covariance operations with fixed-budget conditioning to fit up to 145,008 environmental observations per day.
- Model Risk and Diagnostics:** Compared Matérn and generalized Cauchy dependence models and designed scale- and frequency-resolved tests to identify misspecification in smoothness, noise, and temporal dependence.
- Large-Scale Data Engineering:** Built workflows for missing-data handling, nearest-neighbor construction, dependency-aware ordering, quality filtering, and irregular-grid resampling.
- Estimation and Computation:** Implemented reusable Python/PyTorch pipelines with GLS profiling, L-BFGS optimization, Monte Carlo simulation, CPU/GPU execution, and restartable HPC jobs.

ADDITIONAL

Coursework: Machine Learning, Linear Algebra, Probability Theory, Stochastic Processes, Statistical Computing, Advanced Theory of Statistics I-II, Data Structures and Algorithms, Microeconomics, Econometrics

Languages: Korean (Native); English (Fluent); Chinese (Basic)